

DESIGN CODE OF WALLECE TREE MULTIPLIER

/home/dhruv/cao_lab/vivado_proj/cao_lab_u24ev057/rtl/lab3_wallace.vhd [VHDL]

```
1  -----
2  --
3  -- Company:
4  -- Engineer:
5  --
6  -- Create Date: 09/03/2026 04:51:57 PM
7  -- Design Name:
8  -- Module Name: lab3_wallace - Behavioral
9  -- Project Name:
10 -- Target Devices:
11 -- Tool Versions:
12 -- Description:
13 --
14 -- Dependencies:
15 --
16 -- Revision:
17 -- Revision 0.01 - File Created
18 -- Additional Comments:
19 --
20 -----
21 --
22 library IEEE;
23 use IEEE.STD_LOGIC_1164.ALL;
24
25 -- Uncomment the following library declaration if using
26 -- arithmetic functions with Signed or Unsigned values
27 --use IEEE.NUMERIC_STD.ALL;
28
29 -- Uncomment the following library declaration if instantiating
30 -- any Xilinx leaf cells in this code.
31 --library UNISIM;
32 --use UNISIM.VComponents.all;
33
34 entity lab3_wallace is
35     Port ( a : in STD_LOGIC_VECTOR (3 downto 0);
36           b : in STD_LOGIC_VECTOR (3 downto 0);
37           c : out std_logic_vector (7 downto 0));
38 end lab3_wallace;
39
40 architecture Behavioral of lab3_wallace is
41
42     component rca8 is
43     Port ( a : in STD_LOGIC_VECTOR (7 downto 0);
44           b : in STD_LOGIC_VECTOR (7 downto 0);
45           sum : out STD_LOGIC_VECTOR (7 downto 0);
46           cout : out STD_LOGIC);
```

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47     end component;
48
49     signal p0 : std_logic_vector(7 downto 0);
50     signal p1 : std_logic_vector(7 downto 0);
51     signal p2 : std_logic_vector(7 downto 0);
52     signal p3 : std_logic_vector(7 downto 0);
53
54     signal r0 : std_logic_vector(7 downto 0);
55     signal r1 : std_logic_vector(7 downto 0);
56     signal r2 : std_logic_vector(7 downto 0);
57
58     signal r3 : std_logic_vector(7 downto 0);
59     signal r4 : std_logic_vector(7 downto 0);
60
61     signal cout : std_logic;
62
63 begin
64     process(a , b , p0 , p1 , p2 , p3 , r0 , r1 , r2 , r3 , r4)
65
66     begin
67
68         p0 <= (others => '0');
69         p1 <= (others => '0');
70         p2 <= (others => '0');
71         p3 <= (others => '0');
72
73         for i in 0 to 3 loop
74             p0(i) <= a(0) and b(i);
75             p1(i+1) <= a(1) and b(i);
76             p2(i+2) <= a(2) and b(i);
77             p3(i+3) <= a(3) and b(i);
78         end loop;
79
80         r0 <= (others => '0');
81         r1 <= (others => '0');
82         r2 <= (others => '0');
83         r3 <= (others => '0');
84         r4 <= (others => '0');
85
86         r0(0) <= p0(0);
87         r0(1) <= p0(1) xor p1(1);
88         r0(2) <= p0(2) xor p1(2) xor p2(2);
89
90         r0(3) <= p0(3) xor p1(3) xor p2(3);
91
92         r0(4) <= p3(4) xor p2(4) xor p1(4);
93         r0(5) <= p3(5) xor p2(5);
94         r0(6) <= p3(6);
95         r0(7) <= '0';
96
97

```

98	
99	r2(3) <= p3(3);
100	
101	r1(0) <= '0';
102	r1(1) <= '0';
103	r1(2) <= p0(1) and p1(1);
104	r1(3) <= (p0(2) and p1(2)) or (p1(2) and p2(2)) or (p2(2) and p0(2)
	);
105	
106	r1(4) <= (p0(3) and p1(3)) or (p1(3) and p2(3)) or (p2(3) and p0(3)
	);
107	
108	r1(5) <= (p1(4) and p2(4)) or (p2(4) and p3(4)) or (p3(4) and p1(4)
	);
109	r1(6) <= p2(5) and p3(5);
110	r1(7) <= '0';
111	
112	r3(0) <= r0(0);
113	r3(1) <= r0(1);
114	r3(2) <= r0(2) xor r1(2);
115	r3(3) <= r0(3) xor r1(3) xor r2(3);
116	r3(4) <= r0(4) xor r1(4);
117	r3(5) <= r0(5) xor r1(5);
118	r3(6) <= r0(6) xor r1(6);
119	r3(7) <= r0(7) xor r1(7);
120	
121	r4(0) <= '0';
122	r4(1) <= '0';
123	r4(2) <= '0';
124	r4(3) <= r0(2) and r1(2);
125	r4(4) <= (r0(3) and r1(3)) or (r1(3) and r2(3)) or (r2(3) and r0(3)
	);
126	r4(5) <= r0(4) and r1(4);
127	r4(6) <= r0(5) and r1(5);
128	r4(7) <= r0(6) and r1(6);
129	
130	
131	end process;
132	
133	final_add : rca8 port map (a => r3 , b => r4 , sum => c , cout => cout);
134	
135	end Behavioral;

DESIGN CODE OF RCA

/home/dhruv/cao_lab/vivado_proj/cao_lab_u24ev057/rtl/rca8.vhd [VHDL]	
1	-----
	--
2	-- Company:
3	-- Engineer:
4	--
5	-- Create Date: 08/06/2026 04:21:49 PM

```

6  -- Design Name:
7  -- Module Name: rca8 - Behavioral
8  -- Project Name:
9  -- Target Devices:
10 -- Tool Versions:
11 -- Description:
12 --
13 -- Dependencies:
14 --
15 -- Revision:
16 -- Revision 0.01 - File Created
17 -- Additional Comments:
18 --
19 -----
20 --
21
22 library IEEE;
23 use IEEE.STD_LOGIC_1164.ALL;
24
25 -- Uncomment the following library declaration if using
26 -- arithmetic functions with Signed or Unsigned values
27 --use IEEE.NUMERIC_STD.ALL;
28
29 -- Uncomment the following library declaration if instantiating
30 -- any Xilinx leaf cells in this code.
31 --library UNISIM;
32 --use UNISIM.VComponents.all;
33
34 entity rca8 is
35     Port ( a : in STD_LOGIC_VECTOR (7 downto 0);
36           b : in STD_LOGIC_VECTOR (7 downto 0);
37           sum : out STD_LOGIC_VECTOR (7 downto 0);
38           cout : out STD_LOGIC);
39 end rca8;
40
41 architecture structural of rca8 is
42
43     component ha
44     Port ( A : in STD_LOGIC;
45           B : in STD_LOGIC;
46           SUM : out STD_LOGIC;
47           Cout : out STD_LOGIC
48           );
49     end component;
50
51     component fa
52     Port ( a : in STD_LOGIC;
53           b : in STD_LOGIC;
54           cin : in STD_LOGIC;
55           sum : out STD_LOGIC;
56           cout : out STD_LOGIC);

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57	end component;
58	
59	signal cint : std_logic_vector(6 downto 0);
60	
61	begin
62	
63	inst1 : ha port map(a => a(0) ,b => b(0) ,sum => sum(0) ,cou t => cint(0));
64	inst2 : fa port map(a => a(1) ,b => b(1) ,cin => cint(0) ,sum => sum(1) ,cou t => cint(1));
65	inst3 : fa port map(a => a(2) ,b => b(2) ,cin => cint(1) ,sum => sum(2) ,cou t => cint(2));
66	inst4 : fa port map(a => a(3) ,b => b(3) ,cin => cint(2) ,sum => sum(3) ,cou t => cint(3));
67	inst5 : fa port map(a => a(4) ,b => b(4) ,cin => cint(3) ,sum => sum(4) ,cou t => cint(4));
68	inst6 : fa port map(a => a(5) ,b => b(5) ,cin => cint(4) ,sum => sum(5) ,cou t => cint(5));
69	inst7 : fa port map(a => a(6) ,b => b(6) ,cin => cint(5) ,sum => sum(6) ,cou t => cint(6));
70	inst8 : fa port map(a => a(7) ,b => b(7) ,cin => cint(6) ,sum => sum(7) ,cou t => cout);
71	
72	
73	end structural;

TEST BENCH CODE OF WALLECE TREE MULTIPLIER

/home/dhruv/cao_lab/vivado_proj/cao_lab_u24ev057/tb/lab3_wallace_tb.vhd [VHDL]	
1	-----
	--
2	-- Company:
3	-- Engineer:
4	--
5	-- Create Date: 09/03/2026 05:37:05 PM
6	-- Design Name:
7	-- Module Name: lab3_wallance_tb - Behavioral
8	-- Project Name:
9	-- Target Devices:
10	-- Tool Versions:
11	-- Description:
12	--
13	-- Dependencies:
14	--
15	-- Revision:
16	-- Revision 0.01 - File Created
17	-- Additional Comments:
18	--
19	-----
	--
20	
21	
22	library IEEE;

```

23 use IEEE.STD_LOGIC_1164.ALL;
24
25 -- Uncomment the following library declaration if using
26 -- arithmetic functions with Signed or Unsigned values
27 --use IEEE.NUMERIC_STD.ALL;
28
29 -- Uncomment the following library declaration if instantiating
30 -- any Xilinx leaf cells in this code.
31 --library UNISIM;
32 --use UNISIM.VComponents.all;
33
34 entity lab3_wallance_tb is
35 -- Port ( );
36 end lab3_wallance_tb;
37
38 architecture testing of lab3_wallance_tb is
39
40     component lab3_wallace is
41         Port ( a : in STD_LOGIC_VECTOR (3 downto 0);
42               b : in STD_LOGIC_VECTOR (3 downto 0);
43               c : out STD_LOGIC_VECTOR (7 downto 0));
44     end component;
45
46
47     signal a : std_logic_vector(3 downto 0);
48     signal b : std_logic_vector(3 downto 0);
49     signal c : std_logic_vector(7 downto 0);
50
51 begin
52
53
54     lab3_wallance_test : lab3_wallace port map(
55         a => a,
56         b => b,
57         c => c
58     );
59
60
61     process
62
63     begin
64         a <= "0101";
65         b <= "1010";
66
67         wait for 10ns;
68
69         a <= "0011";
70         b <= "1100";
71
72         wait for 10ns;
73

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74	a <= "1010";
75	b <= "1100";
76	
77	wait for 10ns;
78	
79	a <= "0001";
80	b <= "1000";
81	
82	wait for 10ns;
83	
84	a <= "1110";
85	b <= "1011";
86	
87	wait for 10ns;
88	
89	a <= "1111";
90	b <= "1111";
91	
92	wait for 10ns;
93	
94	a <= "0000";
95	b <= "0000";
96	
97	wait for 10ns;
98	
99	a <= "1010";
100	b <= "1111";
101	
102	wait for 10ns;
103	
104	a <= "0001";
105	b <= "1111";
106	
107	wait for 10ns;
108	
109	a <= "1010";
110	b <= "1111";
111	
112	
113	wait;
114	
115	end process;
116	
117	
118	
119	end testing;

SCREENSHOT OF CODE

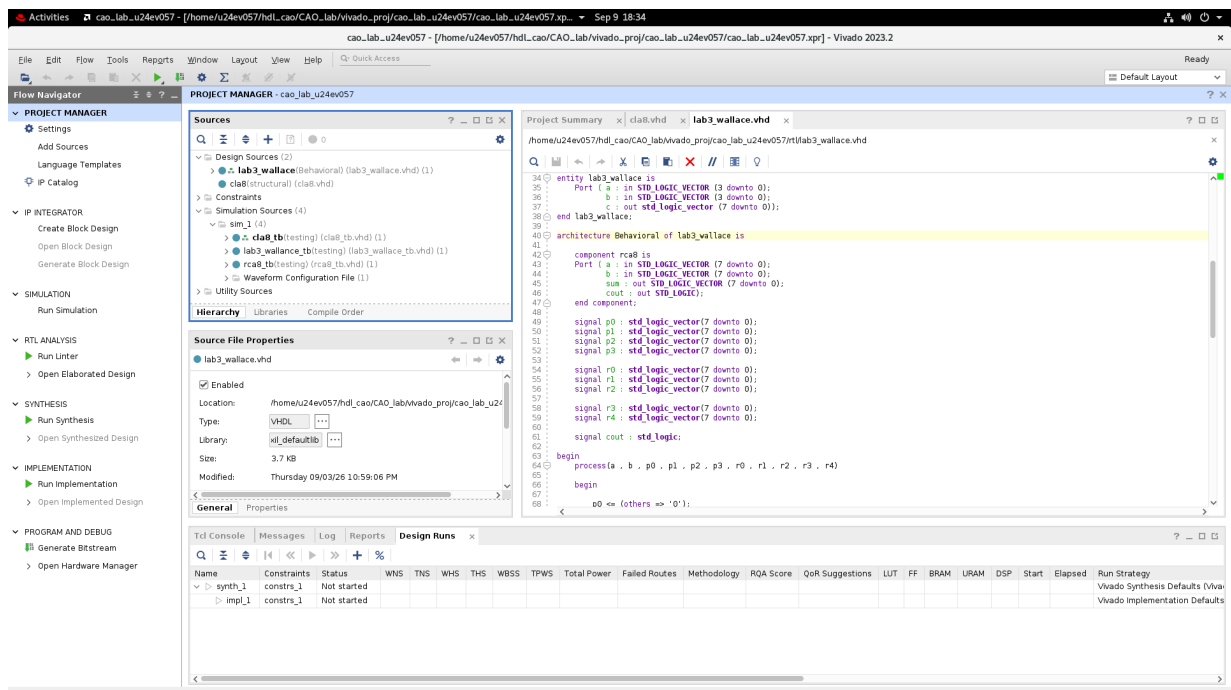


Figure 1: Design

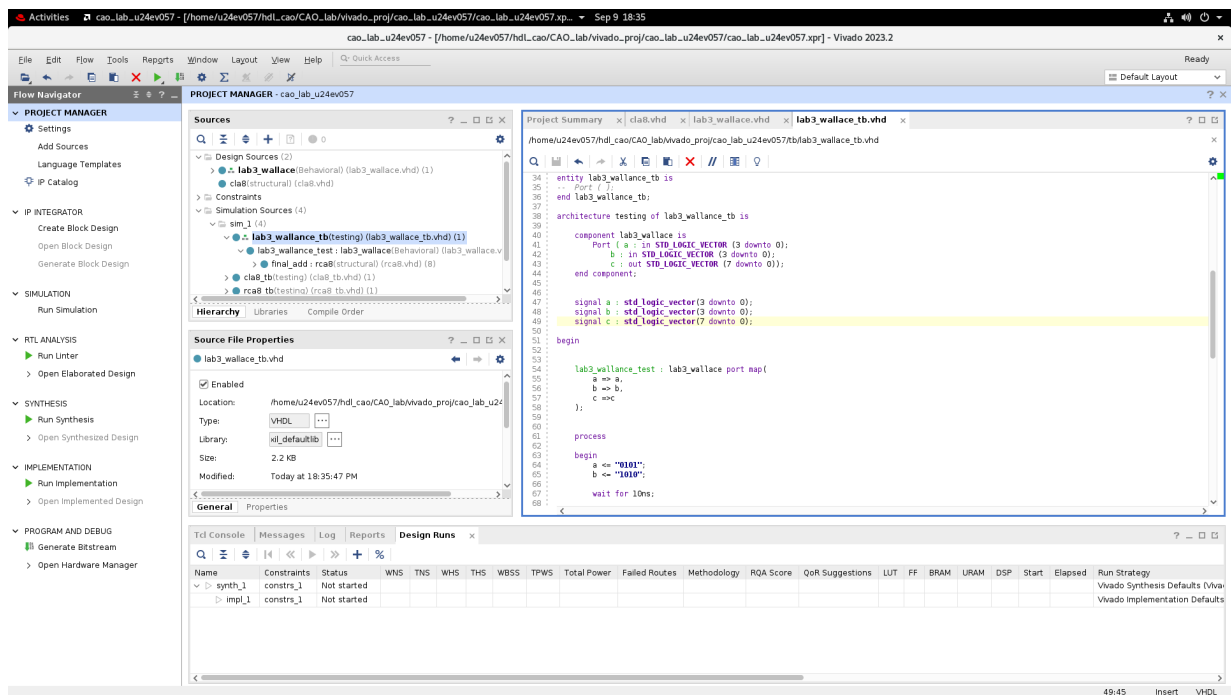


Figure 2: Tb

WAVEFORM

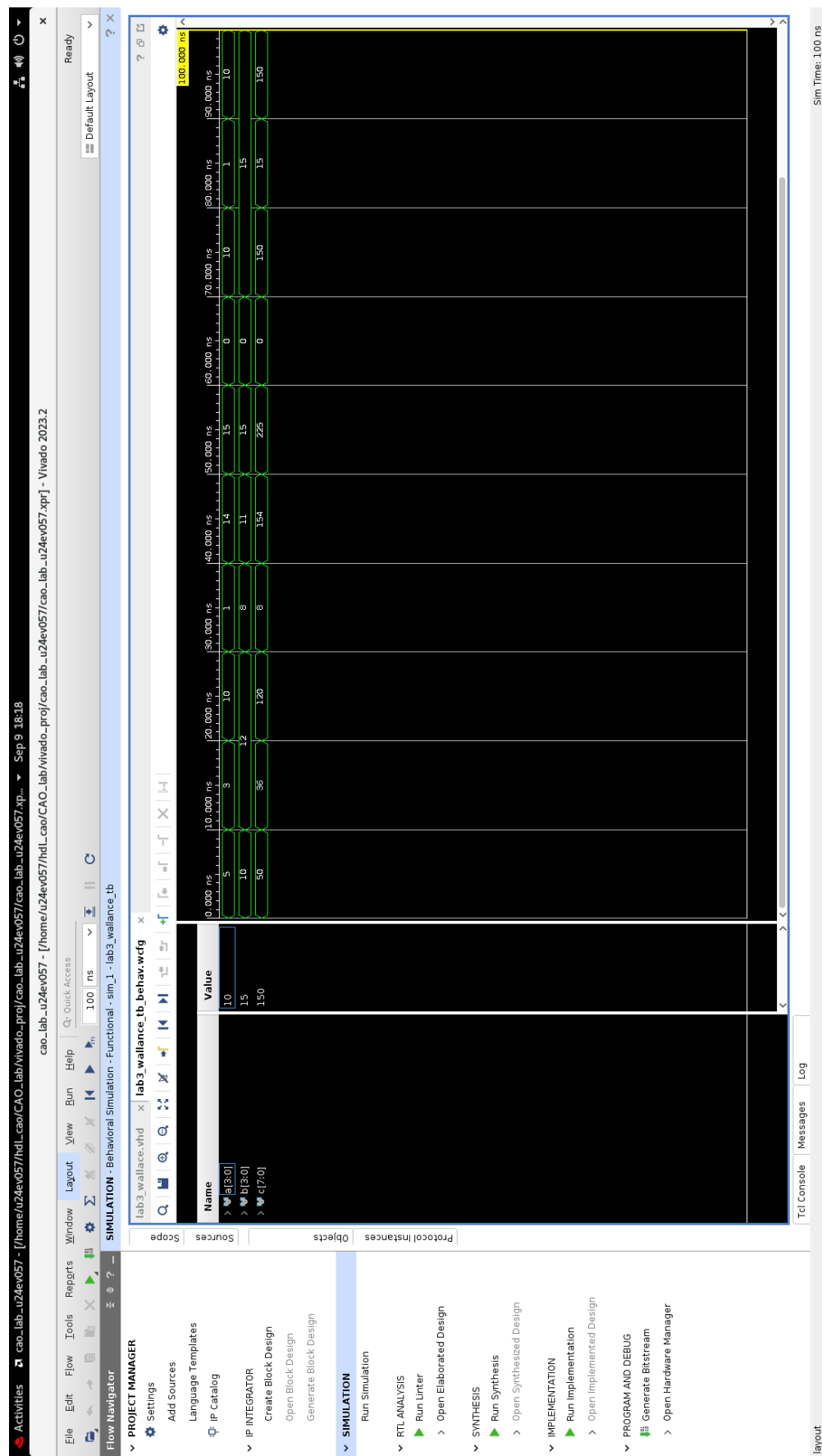


Figure 3: Waveform

